

A scenic mountain landscape with green hills, rocky cliffs, and a cloudy sky. The foreground shows a rocky, grassy slope. In the middle ground, there are steep, rocky cliffs and green hills. The background features more distant, hazy mountain ranges under a sky filled with white and grey clouds.

AI & Software Engineering Workshop

Week 1, Day 4: Memory and State

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The problem

1. Tell your bot: "*my favorite color is blue*"
2. Ask: "*what's my favorite color?*" and it knows ✓
3. Restart the bot. Ask again.
4. **It has no idea who you are.**

The "stateless mode" notice you saw on Day 1 is what we fix today.

Key-value store = lockers

- A wall of lockers 
- **Key** = the locker number → `chat:12345`
- **Value** = whatever you put inside → the conversation
- `get(key)` / `set(key, value)`: that's the whole API
- Ours also has **TTL**: lockers that empty themselves after 30 days

Redis, DynamoDB, memcached, same idea. Ours is SQLite: a database in a single file.

Turn it on

`.env :`

```
SQLITE_PATH=./bot.db
```

- Restart → tell it your favorite color → restart again → ask
- **It remembers.** The file `bot.db` appeared: that's the memory
- No signup, no server, no credit card. It's just a file
- Delete the line → graceful fallback to stateless mode (try it)

Live-code: `/remember` and `/recall`

```
@bot.message_handler(commands=["remember"], func=is_allowed)
def cmd_remember(message):
    note = message.text.split(maxsplit=1)[1] if " " in message.text else ""
    store.set(f"note:{message.from_user.id}", note)
    bot.send_message(message.chat.id, "Saved!")
```

`/recall` is the mirror image: `store.get(f"note:{...}").`

Solo build: a full notes feature

- `/remember <text>` : add a note (not replace!)
- `/recall` : list **all** saved notes
- `/forget` : clear them

Hint: the store saves **strings**. To keep a list, `json.dumps` it on the way in, `json.loads` it on the way out.

Production touches already in the repo

- **Rate limit:** `RATE_LIMIT` env var, default 250 messages/user/day
- **History trimming:** last 20 messages per user, 30-day expiry
- **Whitelist:** `ALLOWED_USERS` makes the bot **silent** to strangers
(silence, not rejection: scanners learn nothing)
- **Dedupe:** Telegram retries webhooks; the bot won't answer twice

All built on the same store you just used.

Debug challenge

The instructor just broke something in your bot.

- Symptom first: what *exactly* stopped working?
- `print()` is a debugger: trace the path of one message
- Found it? Don't fix it yet; explain it to your neighbor first

Today → Tomorrow

Today your bot remembers conversations, notes, and limits, all in one SQLite file.

Tomorrow: the internet. Your bot moves to a real server and answers while your laptop sleeps.